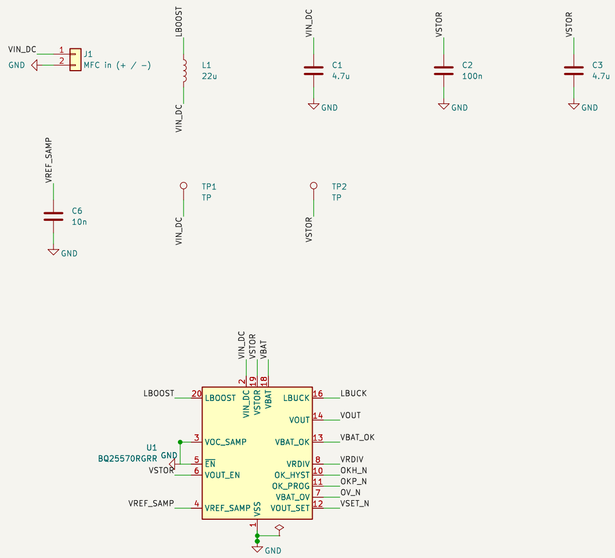
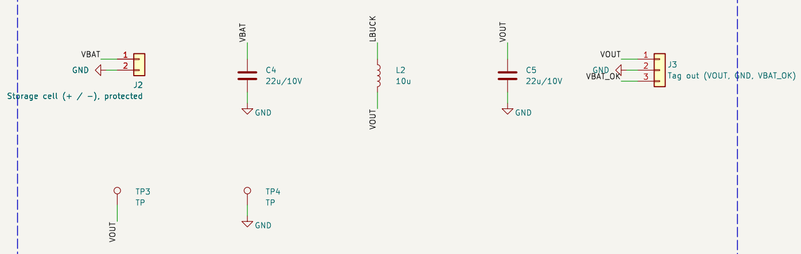
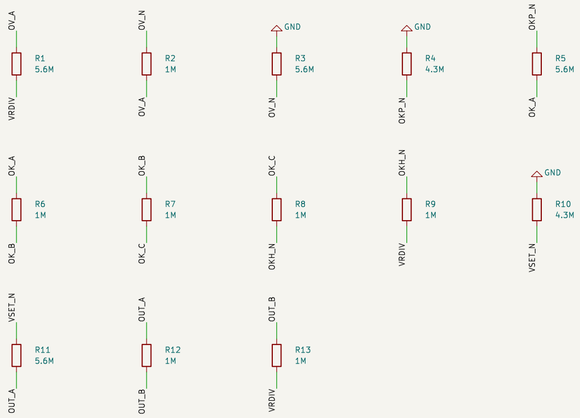
INPUT + HARVESTER: BQ25570 boost charger, MPPT at 50 % of Voc (VOC_SAMP to GND), 22 uH boost inductor



STORAGE + OUTPUT: protected cell, 10 uH buck, 3.3 V rail, VBAT_OK



HIGH-IMPEDANCE SETTING DIVIDERS (no ground pour nearby): VBAT_OV, VBAT_OK / hysteresis, VOUT



BioWatt: BQ25570 values per the TI datasheet (SLUSBH2G) equations. VBAT_OV = 1.5 * VBIAS * (1 + ROV2/ROV1) with VBIAS = 1.21 V: ROV1 = 5.6 M, ROV2 = 5.6 M + 1 M gives 3.95 V (cell charged to about 3.95 V). VBAT_OK = VBIAS * (1 + ROK2/ROK1): ROK1 = 4.3 M, ROK2 = 5.6 M + 1 M + 1 M gives 3.35 V falling; VBAT_OK_HYST = VBIAS * (1 + (ROK2 + ROK3)/ROK1) with ROK3 = 1 M gives 3.63 V rising. VOUT = VBIAS * (1 + ROUT2/ROUT1): ROUT1 = 4.3 M, ROUT2 = 5.6 M + 1 M + 1 M gives 3.35 V. Each divider sum is at most 13 M as the datasheet recommends. Use a PROTECTED single Li-ion/LiPo cell. Clean flux residue thoroughly: the dividers are megohm-level.

First engineering release; see pcb/README.md for status and limits

ClaudeWork

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File: biowatt.kicad_sch

Title: BioWatt MFC energy harvester (BQ25570)

Size: A2	Date: 2026-11-01	**Rev: A**
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 1/1